



Wireless & STP Overview

This note combines guidance on configuring WLANs—both router-based and controller-based deployments—and an introduction to Spanning-Tree Protocol concepts for Layer 2 network redundancy.



Content Sources

- Section 1: Details **WLAN configuration** using wireless routers and WLAN controllers from a 52-page PDF document. ([source](#))
- Section 2: Explains **Spanning-Tree Protocol** fundamentals and Rapid PVST+ operation from a 46-page PDF document. ([source](#))

Section 1: WLAN Configuration & Implementation

This section covers wireless LAN configuration using wireless routers and WLAN controllers from a 52-page PDF document ([source](#))

Module Objective

Implement a **WLAN** using a **wireless router** and a **WLAN Controller (WLC)**, covering remote-site deployment, basic and WPA2-Enterprise WLANs, and troubleshooting.

1. Remote Site WLAN Configuration – The Wireless Router

Key Router Functions

- **Integrated router role:** Combines **wired switch**, **WAN port**, and **wireless radio**
- **Common features:** WLAN security, DHCP server, NAT, QoS, and model-specific options
- **Default login:** Publicly documented defaults - **first security step is changing these**

Basic Router Setup Process

1. **Login:** Enter default IP in browser → use default credentials (commonly *admin/admin*)
2. **Security:** Immediately change administrative password

3. **Network setup**: Modify DHCP IPv4 pool → renew PC IP → re-access with new IP
4. **Wireless configuration**:
 - Review factory SSID, mode, channel, security
 - Choose 802.11 standard (a/b/g/n/ac/ax)
 - Set descriptive **SSID**
 - Select **non-overlapping channel**
 - Configure security mode and passphrase

Coverage Extension

- Single router range: ~45m indoors, ~90m outdoors
- **Mesh networks**: Add APs with same SSID/security but different channels
- Modern vendors provide smartphone apps for automated mesh creation

Advanced Features

- **NAT**: Maps private IPs to single public IP
 - **QoS**: Prioritizes time-sensitive traffic (voice/video)
 - **Port forwarding**: Static rules for specific internal hosts
 - **Port triggering**: Temporary inbound rules after outbound initiation
-

2. Configure a Basic WLAN on the WLC

Network Topology

Device	Interface	IP Address	Subnet Mask
R1	F0/0	172.16.1.1	255.255.255.0
R1	F0/1.1	192.168.200.1	255.255.255.0
WLC	Management	192.168.200.254	255.255.255.0
AP1	Wired 0	192.168.200.3	255.255.255.0

WLC Configuration Steps

1. **Create WLAN**: Define SSID (e.g., *Wireless_LAN*)
 2. **Enable**: Activate WLAN and save settings
 3. **Select Interface**: Choose virtual interface/VLAN
 4. **Security**: Configure Open/WPA/WPA2-PSK
 5. **Verify**: Check operational status under WLANs
 6. **Monitor**: Use Monitor tab for client details (MAC, IP, RSSI)
-

3. Configure a WPA2-Enterprise WLAN on the WLC

Prerequisites

- **SNMP manager** and **RADIUS server** on PC-A
- WLC forwards **SNMP traps** and uses **RADIUS for AAA**

SNMP Configuration

Navigate: **MANAGEMENT → SNMP → Trap Receivers → New**

- Enter **community name** and **IP address**
- Click **Apply**

The screenshot shows the 'SNMP Trap Receiver > New' configuration page in the Cisco WLC Management interface. The left sidebar has a tree view with 'SNMP' expanded, showing options like General, V3 Users, Communities, Trap Receivers, Trap Controls, Trap Logs, HTTP-HTTPS, and IPSEC. The main area contains fields for 'Community Name' (set to 'CCNAv7'), 'IP Address (IPv4/IPv6)' (set to '172.16.1.254'), 'Status' (set to 'Enable'), and 'IPSec' (unchecked). At the top right, there are links for 'Back' and 'Apply'.

RADIUS Configuration

Navigate: **SECURITY → RADIUS → Authentication → New**

- Provide **IP address** of PC-A
- Set **shared secret**
- Click **Apply**

The screenshot shows the 'RADIUS Authentication Servers > New' configuration page in the Cisco WLC Security interface. The left sidebar has a tree view with 'RADIUS' expanded, showing options like General, Authentication, Accounting, Fallback, Downloaded AUP, CACS+, LDAP, Local Root Users, MAC Filtering, Disabled Clients, User Login Policies, AP Policies, Password Policies, Local EAP, Advanced EAP, Priority Order, and Certificate. The main area contains fields for 'Server Index (Priority)' (set to '1'), 'Server IP Address (IPv4/IPv6)' (set to '172.16.1.254'), 'Shared Secret Format' (set to 'ASCII'), 'Shared Secret' (masked with asterisks), 'Confirm Shared Secret' (masked with asterisks), 'Apply Cisco ISE Default settings' (unchecked), 'Key Wrap' (unchecked), 'Port Number' (set to '1812'), 'Server Status' (set to 'Enabled'), 'Support for CoA' (set to 'Disabled'), 'Server Timeout' (set to '5 seconds'), 'Network User' (checked), 'Management' (checked), and 'Management Retransmit Timeout' (set to '5 seconds'). At the top right, there are links for 'Back' and 'Apply'.

VLAN Interface Creation

Navigate: **CONTROLLER → Interfaces → New**

- Name: **vlan5**, ID: **5**
- Assign **physical port** (e.g., G1 → Port 1)
- Set **IP: 192.168.5.254/24**, **Gateway: 192.168.5.1**
- Configure **DHCP server: 192.168.5.1**

DHCP Scope Configuration

Navigate: **Internal DHCP Server → DHCP Scope → New**

- Name: **Wireless_Management**
- Pool: **192.168.200.240-192.168.200.249**
- Default router: **192.168.200.1**
- Enable scope

WPA2-Enterprise WLAN Creation

1. **Create WLAN**: **WLANs → Go**, enter profile name and SSID (*Enterprise_LAN*)
 2. **Enable & Assign**: Enable WLAN, select **vlan5** interface
 3. **Security**: Default **WPA2-AES** with **802.1X**
 4. **AAA Servers**: Select previously configured RADIUS server
-

4. Troubleshooting WLAN Issues

Six-Step Troubleshooting Methodology

1. **Identify** the problem
2. **Establish theory** of probable causes
3. **Test theory** to determine cause
4. **Plan & implement** solution
5. **Verify** full functionality & preventive measures
6. **Document** findings, actions, outcomes

Common Issues & Solutions

Wireless Client Not Connecting

- Check IP settings with ipconfig/ifconfig
- Ping LAN device to test connectivity
- Verify **security mode** and **passphrase** match
- Reload wireless NIC driver if needed

Performance Issues

- Ensure client within **Planned Coverage Area (BSA)**
- Check for **2.4GHz interference**
- Verify **channel selection**

Physical & Link Checks

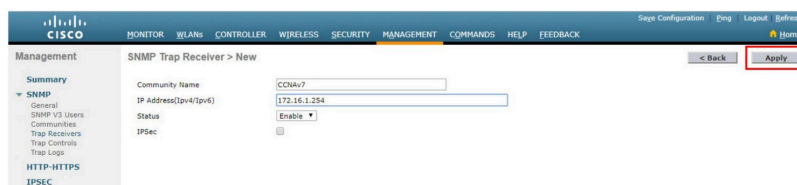
- Confirm power and LED status
- Inspect cables and connections
- Ping AP to verify reachability

Network Optimization Strategies

Strategy	Rationale
Upgrade clients	Avoid 802.11 b/g/n bottlenecks
Split bands	Separate 2.4GHz and 5GHz SSIDs
Improve AP placement	Remove obstructions
Add extenders	Use range extenders or Powerline adapters

Firmware Updates

- Regular updates fix bugs and security issues
- On Cisco 3504 WLC: **WIRELESS → Access Points → Global Configuration → AP Image Pre-download**



5. Key Takeaways

Core Principle: Wireless networking relies on consistent concepts regardless of scale - from home routers to enterprise WLCs.

Wireless Router Fundamentals

- Integrated devices combine switch, WAN, and wireless functions
- Security first:** Change default credentials immediately
- Mesh networks extend coverage beyond single router limits

WLC Essentials

- Lightweight APs** communicate via **LWAPP**
- Centralized management with VLAN-aware design
- Advanced monitoring and configuration capabilities

Enterprise Features

- SNMP** for network monitoring

- **RADIUS** for WPA2-Enterprise authentication
- Systematic troubleshooting using six-step methodology

Best Practices

- Regular firmware updates
 - Band separation for optimal performance
 - Document all configurations and changes
-

Section 2: Spanning-Tree Protocol (STP) Concepts

This section explores STP operations, evolution, and loop prevention mechanisms from a 46-page PDF document ([source](#))

Module Overview

Objective: Explain how **STP enables redundancy** in Layer 2 networks while preventing loops.

1. Purpose of STP – Why Redundancy Needs Loop Prevention

Layer 2 Loop Problems

- **Broadcast storms:** Frames circulate indefinitely
- **MAC table instability:** Constant MAC address changes
- **High CPU utilization:** Network becomes unusable

Key Difference: Layer 3 has TTL/Hop-Limit; Ethernet has **no native TTL**

STP Fundamentals

- **Goal:** Create logical loop-free topology while preserving physical redundancy
 - **Method:** Block selected ports to allow only one logical path
 - **Recalculation:** Unblock ports when links fail
-

2. STP Operations – Building Loop-Free Topology

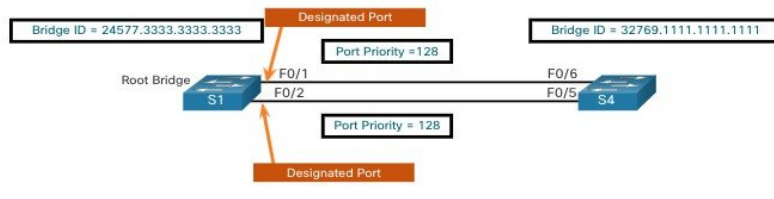
BPDUs Components

- **Bridge Priority:** Default 32768 (range 0-61440 in steps of 4096)
- **Extended System ID:** VLAN identifier
- **MAC Address:** Tie-breaker

Four-Step STA Process

1. **Elect Root Bridge**: Lowest BID wins
2. **Elect Root Ports**: Lowest path cost to root
3. **Elect Designated Ports**: Best path to root per segment
4. **Block Alternate Ports**: Remaining ports blocked

Four-Switch STP Topology



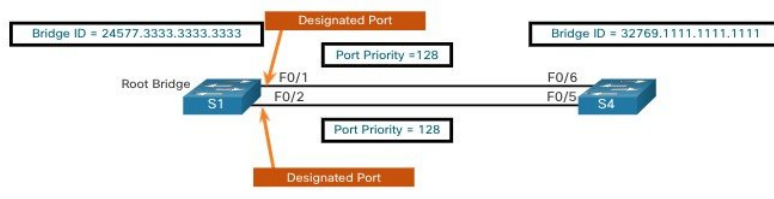
Port Cost Table

Link Speed	STP Cost (802.1D-1998)	RSTP Cost (802.1w-2004)
10 Gbps	2	2,000
1 Gbps	4	20,000
100 Mbps	19	200,000
10 Mbps	100	2,000,000

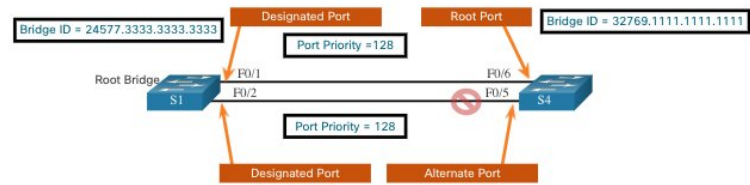
Tie-Breaking Order

1. **Lowest Sender BID**
2. **Lowest Sender Port Priority**
3. **Lowest Sender Port ID**

Two-Switch Example



Detailed Port Role Diagram



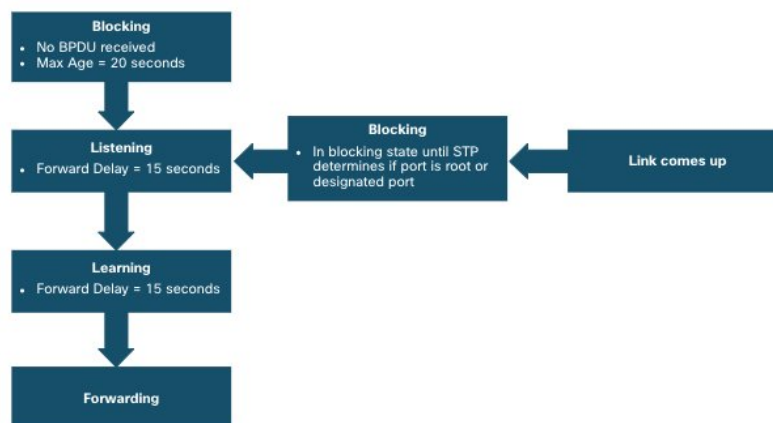
STP Timers

Timer	Default	Function
Hello	2s	BPDU transmission interval
Forward Delay	15s	Listening + Learning duration
Max Age	20s	BPDU validity period

3. STP Port States

State	BPDU	MAC Table	Data
Blocking	Receive	No	No
Listening	Send/Receive	No	No
Learning	Send/Receive	Yes	No
Forwarding	Send/Receive	Yes	Yes
Disabled	None	No	No

STP Port State Diagram



4. STP Evolution – From STP to RSTP

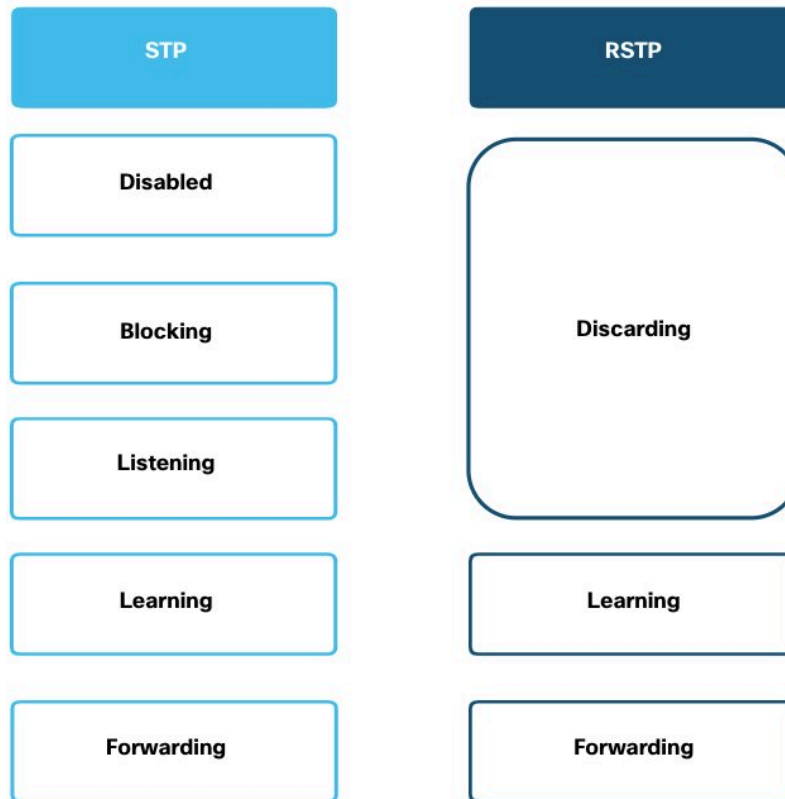
Protocol Varieties

Protocol	Description
STP	Original IEEE 802.1D
PVST+	Cisco per-VLAN STP
RSTP	IEEE 802.1w - faster convergence
Rapid PVST+	Cisco per-VLAN RSTP
MSTP/MST	Multiple spanning tree instances

RSTP Improvements

- **Port States:** Merges Disabled/Blocking/Listening into **Discarding**
- **Port Roles:** Root, Designated, Alternate, Backup
- **Fast Convergence:** Alternate ports transition instantly

STP vs RSTP States



Enhanced Features

- **PortFast**: Bypasses Listening/Learning for access ports
- **BPDU Guard**: Disables PortFast port on BPDU receipt

5. Modern Design Trends

Layer 3 Switching

- Move distribution/core to Layer 3 routing
- Eliminate STP limitations in core
- Keep STP only where Layer 2 required

MSTP Benefits

- Maps multiple VLANs to fewer instances
- Reduces spanning-tree calculations
- Preserves per-instance redundancy

6. Key Takeaways

Core Principle: STP prevents Layer 2 loops while maintaining redundancy through selective port blocking.

Essential Concepts

- **BPDUs** carry Bridge ID for elections
- **Root bridge** has lowest BID
- **Path cost** determines port roles
- **Timers** control convergence speed

Evolution Path

- **STP** → **RSTP** → **Rapid PVST+** → **MSTP**
- Each generation improves convergence time
- Modern networks combine Layer 2/3 designs

Best Practices

- Use **PortFast** only on access ports
- Enable **BPDU Guard** for protection
- Consider **Layer 3** for core redundancy
- Regular **firmware updates** for security

Putting It All Together

Layer-2 Foundations Across Wired and Wireless

Spanning-Tree Protocol (STP) – a mechanism that builds a loop-free logical topology on a switched LAN by blocking selected ports and reconverging when the physical layout changes.

WLAN controller (WLC) – a centralized platform that programs lightweight APs, assigns VLAN-based virtual interfaces, and delegates authentication to external services.

Both technologies operate at **Layer 2**, so the same design principles that keep a wired fabric stable also apply to the wireless fabric:

Aspect	Wired (STP)	Wireless (WLC)
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Redundancy	Blocks backup ports to avoid loops; unblocks on failure.	Deploys multiple APs with identical SSID; traffic is rerouted via the controller when an AP goes down.
Topology awareness	BPDUs propagate bridge IDs and port costs.	LWAPP/DT-CAP packets convey AP capabilities and controller assignments.
Failover speed	RSTP can restore connectivity in sub-second intervals.	Controller can push new WLAN policies instantly to remaining APs.
Management	Root bridge selection, port priorities, and timers are centrally visible.	Global WLAN profile, VLAN interface, SNMP trap receiver, and RADIUS server are configured from a single GUI.

Redundancy & Loop-Prevention in a Unified Network

1. **Physical design** – Place APs on separate switches that participate in STP.
2. **Port roles** –
 - On access-switch ports connecting APs, keep the **designated** role so traffic can flow both ways.
 - Enable **PortFast** only on host ports (PCs, printers) to bypass the 30-second STP delay, preventing DHCP timeouts for wired clients while wireless clients are handled by the controller.
3. **Backup paths** – If a primary AP fails, STP automatically unblocks the alternate link to the backup AP, and the WLC redirects clients to the surviving AP without manual intervention.

Security & Authentication Synergy

- **WPA2-Enterprise** relies on a RADIUS server for 802.1X authentication; the controller forwards authentication requests just as a switch forwards **BPDUs** topology information.
- **BPDUs Guard** on PortFast ports protects against accidental switches being plugged into access ports, complementing **WPA2-Enterprise** by ensuring that rogue devices cannot introduce loops or bypass authentication.

Feature	Wireless focus	Wired focus
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Access control	802.1X via RADIUS, per-SSID policies.	802.1X on switch ports, MAC-based ACLs.
Loop protection	Mesh AP coordination, channel planning.	STP/RSTP port blocking, BPDU Guard.
Monitoring	SNMP traps from WLC, client-RSSI stats.	SNMP traps from switches, STP topology changes.

Operational Best Practices that Span Both Domains

- **Consistent VLAN design** – Assign the same VLAN ID to the wireless management interface (e.g., vlan5) and to the corresponding switch ports; STP uses the VLAN's spanning-tree instance to keep the path stable.
- **Firmware hygiene** – Schedule coordinated upgrades for routers, WLC, APs, and switches; updated STP implementations (RSTP) and WLC images both reduce convergence times and close security gaps.
- **Troubleshooting workflow** – Apply the six-step method universally:
 1. Identify symptom (no Wi-Fi, broadcast storm, high CPU).
 2. List probable causes (wrong SSID, blocked port, mis-configured root bridge).
 3. Test each hypothesis (ping, show spanning-tree, verify RADIUS logs).
 4. Implement fix (adjust channel, change port priority, correct RADIUS secret).
 5. Verify end-to-end connectivity.
 6. Document the change.

Performance Considerations

- **QoS** on the router and switch can prioritize voice/video traffic from both wired hosts and wireless clients; ensure the same class-maps are applied across the L2 domain.
- **Port cost tweaking** – Raising the cost on a high-latency uplink forces STP to prefer a faster path, while the WLC can bias client association to APs on lower-cost links for better throughput.

By aligning **STP's loop-free logic** with the **WLC's centralized WLAN management**, a network engineer can craft a resilient, secure environment where wired and wireless segments cooperate seamlessly, recover quickly from failures, and maintain consistent policy enforcement across the entire Layer 2 fabric.



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Implement a **WLAN** using a **wireless router** and a **WLAN Controller (WLC)**, covering remote-site deployment, basic and WPA2-Enterprise WLANs, and troubleshooting.

1. Remote Site WLAN Configuration – The Wireless Router

Key Router Functions

- **Integrated router role**: Combines **wired switch**, **WAN port**, and **wireless radio**
- **Common features**: WLAN security, DHCP server, NAT, QoS, and model-specific options
- **Default login**: Publicly documented defaults - **first security step is changing these**

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1. **Login**: Enter default IP in browser → use default credentials (commonly *admin/admin*)
2. **Security**: Immediately change administrative password

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4. **Wireless configuration**:
 - Review factory SSID, mode, channel, security
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Advanced Features

- **NAT**: Maps private IPs to single public IP
 - **QoS**: Prioritizes time-sensitive traffic (voice/video)
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2. Configure a Basic WLAN on the WLC

Network Topology

Device	Interface	IP Address	Subnet Mask
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R1	F0/1.1	192.168.200.1	255.255.255.0
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AP1	Wired 0	192.168.200.3	255.255.255.0

WLC Configuration Steps

1. **Create WLAN**: Define SSID (e.g., *Wireless_LAN*)
 2. **Enable**: Activate WLAN and save settings
 3. **Select Interface**: Choose virtual interface/VLAN
 4. **Security**: Configure Open/WPA/WPA2-PSK
 5. **Verify**: Check operational status under WLANs
 6. **Monitor**: Use Monitor tab for client details (MAC, IP, RSSI)
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3. Configure a WPA2-Enterprise WLAN on the WLC

Prerequisites

- **SNMP manager** and **RADIUS server** on PC-A
- WLC forwards **SNMP traps** and uses **RADIUS for AAA**

SNMP Configuration

Navigate: **MANAGEMENT → SNMP → Trap Receivers → New**

- Enter **community name** and **IP address**
- Click **Apply**

The screenshot shows the 'SNMP Trap Receiver > New' configuration page in the Cisco WLC Management interface. The left sidebar shows the navigation tree with 'SNMP' expanded. The main area contains the following fields:

Field	Value
Community Name	CCNAv7
IP Address (IPv4/IPv6)	172.16.1.254
Status	Enable
IPSec	☐

Buttons for '< Back' and 'Apply' are visible at the top right.

RADIUS Configuration

Navigate: **SECURITY → RADIUS → Authentication → New**

- Provide **IP address** of PC-A
- Set **shared secret**
- Click **Apply**

The screenshot shows the 'RADIUS Authentication Servers > New' configuration page in the Cisco WLC Security interface. The left sidebar shows the navigation tree with 'RADIUS' expanded. The main area contains the following fields:

Field	Value
Server Index (Priority)	1
Server IP Address (IPv4/IPv6)	172.16.1.254
Shared Secret Format	ASCII
Shared Secret	*****
Confirm Shared Secret	*****
Apply Cisco ISE Default settings	☐
Key Wrap	☐ (Designed for FIPS customers and requires a key wrap compliant RADIUS server)
Port Number	1812
Server Status	Enabled
Support for CoA	Disabled
Server Timeout	5 seconds
Network User	☒ Enable
Management	☒ Enable
Management Retransmit Timeout	5 seconds

Buttons for '< Back' and 'Apply' are visible at the top right.

VLAN Interface Creation

Navigate: **CONTROLLER → Interfaces → New**

- Name: **vlan5**, ID: **5**
- Assign **physical port** (e.g., G1 → Port 1)
- Set **IP: 192.168.5.254/24**, **Gateway: 192.168.5.1**
- Configure **DHCP server: 192.168.5.1**

DHCP Scope Configuration

Navigate: **Internal DHCP Server → DHCP Scope → New**

- Name: **Wireless_Management**
- Pool: **192.168.200.240-192.168.200.249**
- Default router: **192.168.200.1**
- Enable scope

WPA2-Enterprise WLAN Creation

1. **Create WLAN**: **WLANs → Go**, enter profile name and SSID (*Enterprise_LAN*)
 2. **Enable & Assign**: Enable WLAN, select **vlan5** interface
 3. **Security**: Default **WPA2-AES** with **802.1X**
 4. **AAA Servers**: Select previously configured RADIUS server
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4. Troubleshooting WLAN Issues

Six-Step Troubleshooting Methodology

1. **Identify** the problem
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- Check IP settings with ipconfig/ifconfig
- Ping LAN device to test connectivity
- Verify **security mode** and **passphrase** match
- Reload wireless NIC driver if needed

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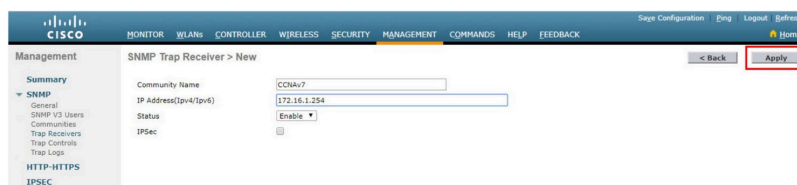
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Split bands	Separate 2.4GHz and 5GHz SSIDs
Improve AP placement	Remove obstructions
Add extenders	Use range extenders or Powerline adapters

Firmware Updates

- Regular updates fix bugs and security issues
- On Cisco 3504 WLC: **WIRELESS → Access Points → Global Configuration → AP Image Pre-download**



5. Key Takeaways

Core Principle: Wireless networking relies on consistent concepts regardless of scale - from home routers to enterprise WLCs.

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Enterprise Features

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Best Practices

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 - Band separation for optimal performance
 - Document all configurations and changes
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Section 2: Spanning-Tree Protocol (STP) Concepts

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1. Purpose of STP – Why Redundancy Needs Loop Prevention

Layer 2 Loop Problems

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Key Difference: Layer 3 has TTL/Hop-Limit; Ethernet has **no native TTL**

STP Fundamentals

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 - **Method:** Block selected ports to allow only one logical path
 - **Recalculation:** Unblock ports when links fail
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2. STP Operations – Building Loop-Free Topology

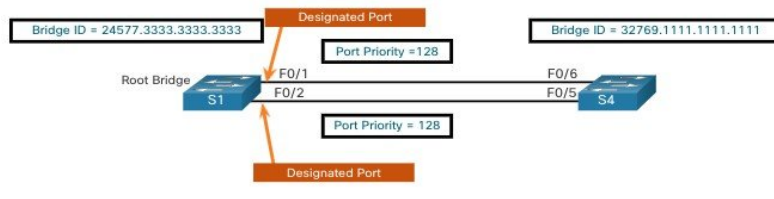
BPDU Components

- **Bridge Priority:** Default 32768 (range 0-61440 in steps of 4096)
- **Extended System ID:** VLAN identifier
- **MAC Address:** Tie-breaker

Four-Step STA Process

1. **Elect Root Bridge**: Lowest BID wins
2. **Elect Root Ports**: Lowest path cost to root
3. **Elect Designated Ports**: Best path to root per segment
4. **Block Alternate Ports**: Remaining ports blocked

Four-Switch STP Topology



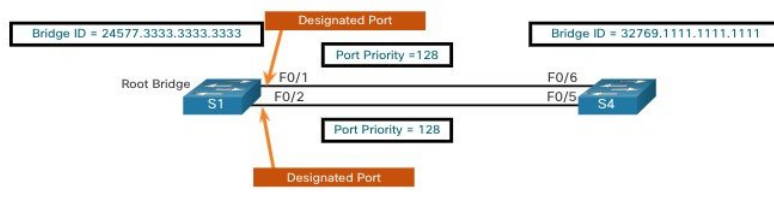
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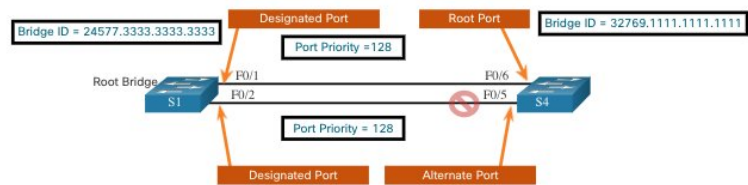
Tie-Breaking Order

1. **Lowest Sender BID**
2. **Lowest Sender Port Priority**
3. **Lowest Sender Port ID**

Two-Switch Example



Detailed Port Role Diagram



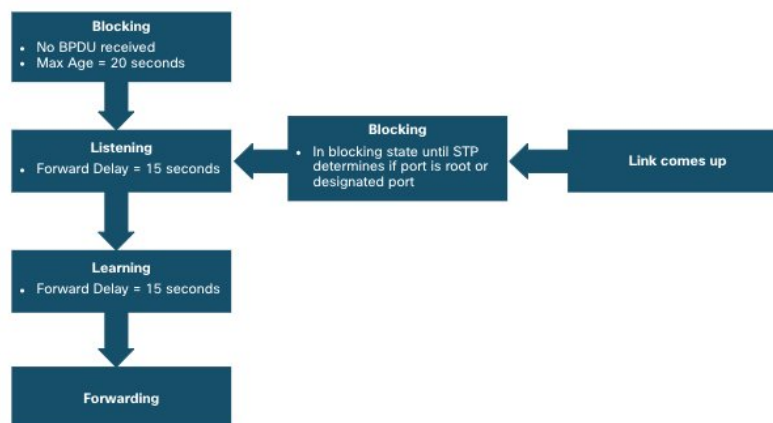
STP Timers

Timer	Default	Function
Hello	2s	BPDUs transmission interval
Forward Delay	15s	Listening + Learning duration
Max Age	20s	BPDUs validity period

3. STP Port States

State	BPDUs	MAC Table	Data
Blocking	Receive	No	No
Listening	Send/Receive	No	No
Learning	Send/Receive	Yes	No
Forwarding	Send/Receive	Yes	Yes
Disabled	None	No	No

STP Port State Diagram



4. STP Evolution – From STP to RSTP

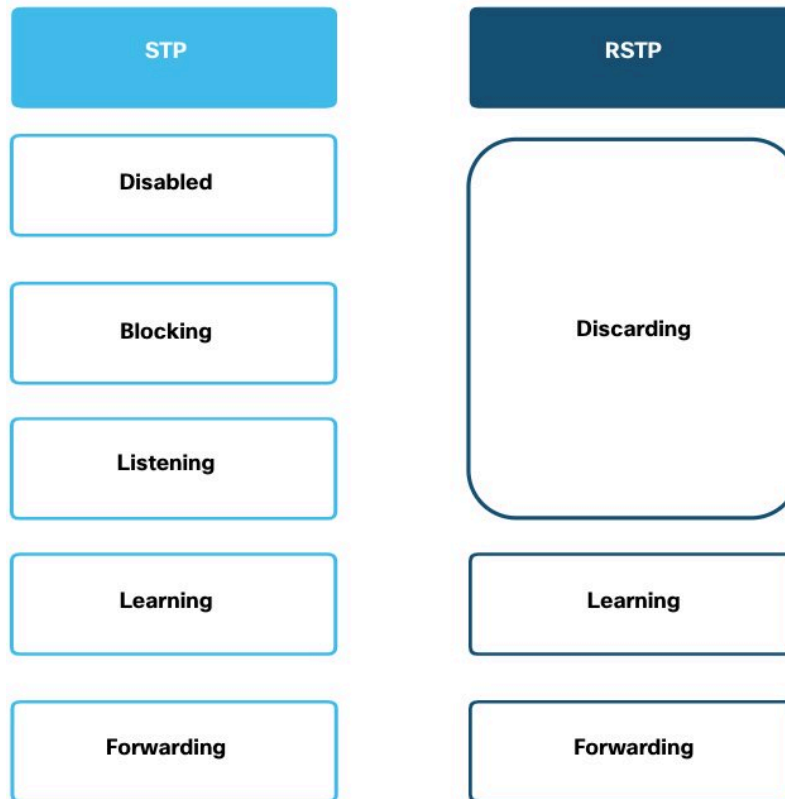
Protocol Varieties

Protocol	Description
STP	Original IEEE 802.1D
PVST+	Cisco per-VLAN STP
RSTP	IEEE 802.1w - faster convergence
Rapid PVST+	Cisco per-VLAN RSTP
MSTP/MST	Multiple spanning tree instances

RSTP Improvements

- **Port States:** Merges Disabled/Blocking/Listening into **Discarding**
- **Port Roles:** Root, Designated, Alternate, Backup
- **Fast Convergence:** Alternate ports transition instantly

STP vs RSTP States



Enhanced Features

- **PortFast**: Bypasses Listening/Learning for access ports
- **BPDU Guard**: Disables PortFast port on BPDU receipt

5. Modern Design Trends

Layer 3 Switching

- Move distribution/core to Layer 3 routing
- Eliminate STP limitations in core
- Keep STP only where Layer 2 required

MSTP Benefits

- Maps multiple VLANs to fewer instances
- Reduces spanning-tree calculations
- Preserves per-instance redundancy

6. Key Takeaways

Core Principle: STP prevents Layer 2 loops while maintaining redundancy through selective port blocking.

Essential Concepts

- **BPDUs** carry Bridge ID for elections
- **Root bridge** has lowest BID
- **Path cost** determines port roles
- **Timers** control convergence speed

Evolution Path

- **STP** → **RSTP** → **Rapid PVST+** → **MSTP**
- Each generation improves convergence time
- Modern networks combine Layer 2/3 designs

Best Practices

- Use **PortFast** only on access ports
- Enable **BPDU Guard** for protection
- Consider **Layer 3** for core redundancy
- Regular **firmware updates** for security

Putting It All Together

Layer-2 Foundations Across Wired and Wireless

Spanning-Tree Protocol (STP) – a mechanism that builds a loop-free logical topology on a switched LAN by blocking selected ports and reconverging when the physical layout changes.

WLAN controller (WLC) – a centralized platform that programs lightweight APs, assigns VLAN-based virtual interfaces, and delegates authentication to external services.

Both technologies operate at **Layer 2**, so the same design principles that keep a wired fabric stable also apply to the wireless fabric:

Aspect	Wired (STP)	Wireless (WLC)
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Redundancy	Blocks backup ports to avoid loops; unblocks on failure.	Deploys multiple APs with identical SSID; traffic is rerouted via the controller when an AP goes down.
Topology awareness	BPDUs propagate bridge IDs and port costs.	LWAPP/DT-CAP packets convey AP capabilities and controller assignments.
Failover speed	RSTP can restore connectivity in sub-second intervals.	Controller can push new WLAN policies instantly to remaining APs.
Management	Root bridge selection, port priorities, and timers are centrally visible.	Global WLAN profile, VLAN interface, SNMP trap receiver, and RADIUS server are configured from a single GUI.



Redundancy & Loop-Prevention in a Unified Network

1. **Physical design** – Place APs on separate switches that participate in STP.
2. **Port roles** –
 - On access-switch ports connecting APs, keep the **designated** role so traffic can flow both ways.
 - Enable **PortFast** only on host ports (PCs, printers) to bypass the 30-second STP delay, preventing DHCP timeouts for wired clients while wireless clients are handled by the controller.
3. **Backup paths** – If a primary AP fails, STP automatically unblocks the alternate link to the backup AP, and the WLC redirects clients to the surviving AP without manual intervention.



Security & Authentication Synergy

- **WPA2-Enterprise** relies on a RADIUS server for 802.1X authentication; the controller forwards authentication requests just as a switch forwards **BPDUs** topology information.
- **BPDUs Guard** on PortFast ports protects against accidental switches being plugged into access ports, complementing **WPA2-Enterprise** by ensuring that rogue devices cannot introduce loops or bypass authentication.

Feature	Wireless focus	Wired focus
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Access control	802.1X via RADIUS, per-SSID policies.	802.1X on switch ports, MAC-based ACLs.
Loop protection	Mesh AP coordination, channel planning.	STP/RSTP port blocking, BPDU Guard.
Monitoring	SNMP traps from WLC, client-RSSI stats.	SNMP traps from switches, STP topology changes.

Operational Best Practices that Span Both Domains

- **Consistent VLAN design** – Assign the same VLAN ID to the wireless management interface (e.g., vlan5) and to the corresponding switch ports; STP uses the VLAN's spanning-tree instance to keep the path stable.
- **Firmware hygiene** – Schedule coordinated upgrades for routers, WLC, APs, and switches; updated STP implementations (RSTP) and WLC images both reduce convergence times and close security gaps.
- **Troubleshooting workflow** – Apply the six-step method universally:
 1. Identify symptom (no Wi-Fi, broadcast storm, high CPU).
 2. List probable causes (wrong SSID, blocked port, mis-configured root bridge).
 3. Test each hypothesis (ping, show spanning-tree, verify RADIUS logs).
 4. Implement fix (adjust channel, change port priority, correct RADIUS secret).
 5. Verify end-to-end connectivity.
 6. Document the change.

Performance Considerations

- **QoS** on the router and switch can prioritize voice/video traffic from both wired hosts and wireless clients; ensure the same class-maps are applied across the L2 domain.
- **Port cost tweaking** – Raising the cost on a high-latency uplink forces STP to prefer a faster path, while the WLC can bias client association to APs on lower-cost links for better throughput.

By aligning **STP's loop-free logic** with the **WLC's centralized WLAN management**, a network engineer can craft a resilient, secure environment where wired and wireless segments cooperate seamlessly, recover quickly from failures, and maintain consistent policy enforcement across the entire Layer 2 fabric.